In [45]:
```python
import pandas as pd

df = pd.read_csv("mallcustomers.csv")
```

In [46]:
```python
print(df.head())
```

```
   CustomerID  Gender  Age      Income  SpendingScore
0           1    Male   19  15,000 USD             39
1           2    Male   21  15,000 USD             81
2           3  Female   20  16,000 USD              6
3           4  Female   23  16,000 USD             77
4           5  Female   31  17,000 USD             40
```

In [47]:
```python
print(df.dtypes)
```

```
CustomerID        int64
Gender           object
Age               int64
Income           object
SpendingScore     int64
dtype: object
```

In [48]:
```python
# Convert Income from string to numeric
df["Income"] = (
    df["Income"]
      .str.replace(",", "", regex=False)      # Remove commas
      .str.replace(" USD", "", regex=False)   # Remove " USD"
      .astype(int)                            # Convert to integer
)
```

In [49]:
```python
print(df.dtypes)
```

```
CustomerID        int64
Gender           object
Age               int64
Income            int64
SpendingScore     int64
dtype: object
```

In [50]:
```python
print(df.head())
```

```
   CustomerID  Gender  Age  Income  SpendingScore
0           1    Male   19   15000             39
1           2    Male   21   15000             81
2           3  Female   20   16000              6
3           4  Female   23   16000             77
4           5  Female   31   17000             40
```

In [51]:
```python
df = df.drop(columns=["CustomerID"])
```

In [52]:
```python
print(df.head())
```

```
   Gender  Age  Income  SpendingScore
0    Male   19   15000             39
1    Male   21   15000             81
2  Female   20   16000              6
3  Female   23   16000             77
4  Female   31   17000             40
```

In [53]:
```python
print(df.columns)
```

```
Index(['Gender', 'Age', 'Income', 'SpendingScore'], dtype='object')
```

In [54]:
```python
print(df[["Income", "SpendingScore"]].describe())
```

```
              Income  SpendingScore
count     200.000000     200.000000
mean    60560.000000      50.200000
std     26264.721165      25.823522
min     15000.000000       1.000000
25%     41500.000000      34.750000
50%     61500.000000      50.000000
75%     78000.000000      73.000000
max    137000.000000      99.000000
```

In [55]:
```python
from sklearn.preprocessing import StandardScaler

# Select only the features used for clustering
X = df[["Income", "SpendingScore"]]
```

```python
# Create the scaler
scaler = StandardScaler()

# Normalize the data
X_scaled = scaler.fit_transform(X)

# Convert back to a DataFrame for easier viewing
X_scaled = pd.DataFrame(X_scaled, columns=["Income", "SpendingScore"])

# Display the first few rows
print(X_scaled.head())
```

```
     Income  SpendingScore
0 -1.738999      -0.434801
1 -1.738999       1.195704
2 -1.700830      -1.715913
3 -1.700830       1.040418
4 -1.662660      -0.395980
```

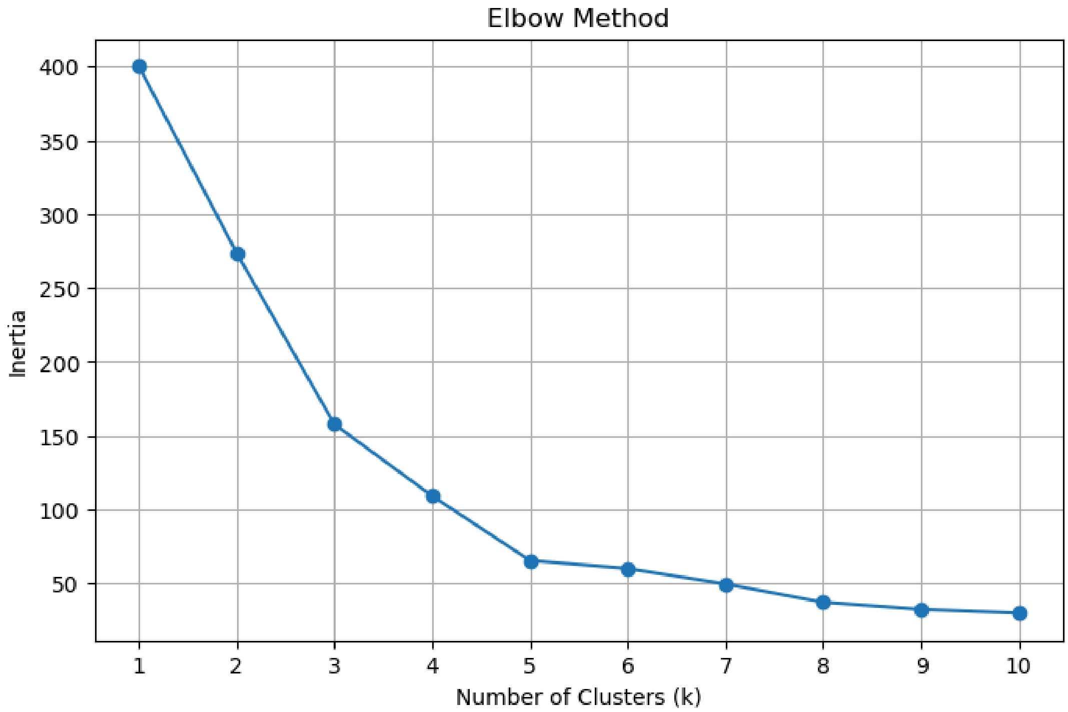
In [56]:

```python
print(X_scaled.describe())
```

```
             Income  SpendingScore
count  2.000000e+02   2.000000e+02
mean  -1.421085e-16  -1.465494e-16
std    1.002509e+00   1.002509e+00
min   -1.738999e+00  -1.910021e+00
25%   -7.275093e-01  -5.997931e-01
50%    3.587926e-02  -7.764312e-03
75%    6.656748e-01   8.851316e-01
max    2.917671e+00   1.894492e+00
```

In [57]:

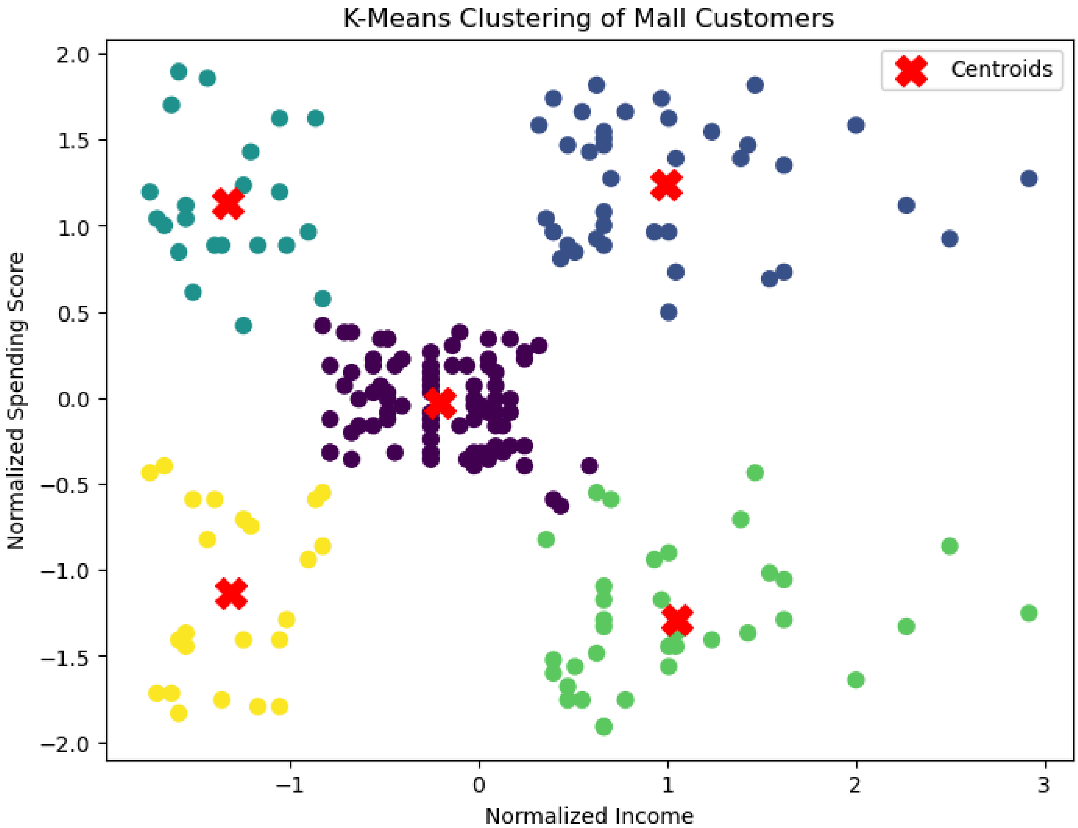
```python
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

# Calculate inertia for k = 1 through 10
inertia = []

for k in range(1, 11):
    kmeans = KMeans(n_clusters=k, random_state=42)
    kmeans.fit(X_scaled)
    inertia.append(kmeans.inertia_)

# Plot the Elbow Method
plt.figure(figsize=(8, 5))
plt.plot(range(1, 11), inertia, marker='o')
plt.title("Elbow Method")
plt.xlabel("Number of Clusters (k)")
plt.ylabel("Inertia")
plt.xticks(range(1, 11))
plt.grid(True)
plt.show()
```



In [29]:
```
k = 5
```

In [31]:
```
# Train the final model
kmeans = KMeans(n_clusters=5, random_state=42)

# Create cluster assignments
clusters = kmeans.fit_predict(X_scaled)

# Add cluster labels to the dataframe
df["Cluster"] = clusters
```

In [33]:
```
plt.figure(figsize=(8, 6))

plt.scatter(
    X_scaled["Income"],
    X_scaled["SpendingScore"],
    c=clusters,
    cmap="viridis",
    s=50
)

# Plot the cluster centers
plt.scatter(
    kmeans.cluster_centers_[:, 0],
    kmeans.cluster_centers_[:, 1],
    c="red",
    marker="X",
    s=200,
    label="Centroids"
)

plt.title("K-Means Clustering of Mall Customers")
plt.xlabel("Normalized Income")
plt.ylabel("Normalized Spending Score")
plt.legend()
plt.show()
```



In [35]:
```
#The Elbow Method was used to find the best number of clusters by testing different values of k from 1 to 10
# and comparing the inertia values. Based on the elbow plot, k = 5 appeared to be the best choice because it
# showed a noticeable point where adding more clusters did not significantly improve the results. A final
# K-Means model was created using 5 clusters and `random_state=42` to make sure the results could be reproduced.
# The scatter plot shows the five customer groups and the location of each cluster’s centroid.
```

In [37]:
```
# Create dummy variables for Gender
df = pd.get_dummies(df, columns=["Gender"])

# Check the new columns
print(df.head())
```

```
   Age  Income  SpendingScore  Cluster  Gender_Female  Gender_Male
0   19   15000             39        4          False         True
1   21   15000             81        2          False         True
2   20   16000              6        4           True        False
3   23   16000             77        2           True        False
4   31   17000             40        4           True        False
```

In [39]:
```
# Add cluster labels to the original dataframe
df["Cluster"] = clusters
```

In [41]:
```
# Calculate mean age and gender distribution by cluster
cluster_summary = df.groupby("Cluster").agg(
    Mean_Age=("Age", "mean"),
    Female_Count=("Gender_Female", "sum"),
    Male_Count=("Gender_Male", "sum"),
    Total_Customers=("Cluster", "count")
)

# Calculate gender percentages
cluster_summary["Female_Percentage"] = (
    cluster_summary["Female_Count"] / cluster_summary["Total_Customers"] * 100
)

cluster_summary["Male_Percentage"] = (
    cluster_summary["Male_Count"] / cluster_summary["Total_Customers"] * 100
)

cluster_summary
```

Out[41]:

Cluster	Mean_Age	Female_Count	Male_Count	Total_Customers	Female_Percentage	Male_Percentage
0	42.716049	48	33	81	59.259259	40.740741
1	32.692308	21	18	39	53.846154	46.153846
2	25.272727	13	9	22	59.090909	40.909091
3	41.114286	16	19	35	45.714286	54.285714
4	45.217391	14	9	23	60.869565	39.130435

In [43]:

```
# After looking at the clusters and adding the demographic information, it was clear that each group
# had different shopping behaviors. The low income and low spending cluster appears to represent customers
# who spend less and may be more careful with their purchases. The high income and high spending cluster
# represents customers who have more spending power and may be some of the mall’s most valuable customers.
# The low income and high spending cluster may include customers who enjoy shopping even though they have
# lower incomes, while the high income and low spending cluster may represent customers who earn more but do
# not spend as much at the mall.

#Looking at the gender distribution and average age for each cluster provided more insight into the different
# customer groups. This information helps mall management better understand who their customers are and can
# help them create more effective marketing strategies based on each group’s characteristics.
```

In [ ]:

```
# **Memo to: Mall Management**
# **Subject: Customer Segmentation Analysis Using K-Means Clustering**

# The purpose of this analysis was to identify different customer segments based on income and spending
# habits using the mall customer dataset. A K-Means clustering model was used to group customers with similar
# characteristics and provide insight into customer behavior. Before creating the model, the data was
# prepared by reviewing the dataset and converting the Income variable from a text format into a numeric format.
# The CustomerID variable was removed because it was only an identifier and did not provide useful information
# for customer segmentation. Since Income and SpendingScore were measured on different scales, z-score
# normalization was applied to standardize the data and ensure that both variables contributed equally to the
# clustering process. The Elbow Method was used to determine the best number of clusters for the K-Means model.
# After comparing different values of k, the elbow plot showed that 5 clusters provided the best grouping of
# customers. A K-Means model was then created using 5 clusters with `random_state=42` to ensure that the results
# could be reproduced. The five customer groups were analyzed based on their locations in the clustering
# visualization. The clusters represented different customer behaviors, including customers with lower income
# and lower spending, customers with higher income and higher spending, and other groups with different
# combinations of income and spending patterns. These groups provide insight into how customers differ in their
# shopping habits. Additional demographic information was analyzed by using Gender and Age. Dummy variables
# were created for Male and Female customers, and the average age and gender distribution were calculated for
# each cluster. This additional information helped provide a better understanding of the characteristics of
# each customer segment. Overall, the clustering analysis shows that customers can be separated into distinct
# groups based on income and spending behavior. These findings can help Mall Management better understand
# their customers and develop more targeted marketing strategies to better meet the needs of each customer group.
```